

Relationship among process parameters, microstructure, and mechanical properties of austempered ductile iron (ADI)

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ABSTRACT

The study investigated the relationship among process parameters, microstructure, and mechanical properties of austempered ductile cast iron (ADI). ADI was composed of spheroidal graphite, acicular ferrite, and high-carbon austenite. Austempering process parameters affected the microstructure including the ferrite size, the volume fraction of austenite and its carbon concentration, and resulted in varying degrees of mechanical response. High carbon concentration in prior austenite obtained at high austenitizing temperature improved its strength, resulting in the formation of thin acicular ferrite. The high-carbon austenite and acicular ferrites of ADI were coarsened under increased austempering temperature. The austempering time affected the microstructure significantly at durations less than 60 min but affected little when the holding time was further extended. ADI exhibited a strong combination of mechanical properties with an ultimate tensile strength (UTS) of 1348 MPa and impact energy of 70 J when austenitized at 900 °C for 90 min and austempered at 300 °C for 60 min. High strength was mainly attributed to fine acicular ferrite, and the robustness was related to fine microstructure with appropriate levels of high-carbon austenite.

1. Introduction

Austempered ductile cast iron (ADI) is an important engineering material of great industrial interest because of its excellent mechanical properties and low production costs [1,2]. Additionally, ADI is a promising self-lubricant material due to the presence of spheroidal graphite, which contributes to the widespread commercial application of ADI in structural components used in transmission [3,4]. Currently, ADI has partially replaced forged steel in agricultural machinery, automotive, military, and other industries [5,6].

The production of ADI [7] entails heating of ductile iron first to an austenitizing temperature to form a complete austenitic matrix, followed by cooling to an austempering temperature for a specific duration to generate an ausferrite matrix consisting of acicular ferrite (α) and high-carbon austenite (γ) [8]. The heat treatment parameters play a key role in the microstructure and mechanical properties of ADI [9]. The strength of ADI exceeds 1400 MPa when the austempering temperature is 230 °C, but the strength is only 900 MPa when the austempering temperature is 380 °C [4]. Previous studies mostly focused on strength improvement. Enhanced safety criteria for engineering materials emphasize the need for toughness [10,11]. Although ADI exhibits

strength comparable to that of forged steels, the ductility is very low [12]. The ductility of ADI is improved by 50% through partitioning treatment [13]. However, the toughness of ADI does not meet the criteria specified for transmission parts.

The effects of temperature and time of austenitizing and austempering on ADI have been reported previously [14,15]. However, the existed investigations only consider one parameter, thus the results are not systematic and some results reported by different authors are even contradictory. For example, Susil K et al. [16] and M. Górny et al. [17] studied the effect of austenitizing temperature on the microstructure of ADI, but obvious different microstructures were observed; ADI austempered at similar austempering temperature exhibited different microstructure and mechanical responses [4,18]. Moreover, the austenitizing temperature of ADI is commonly in the range from 850 °C to 950 °C in previous investigations [16,17]. However, the spheroidal graphite in ductile iron can provide enough carbon atoms during austenization treatment, thus carbon concentration in austenite would increase with the increase in austenitizing temperature according to the Fe-C-Si phase diagram [19]. Meanwhile, carbon concentration in austenite directly determines the transformation kinetics at subsequent austempering treatment. Therefore, the austenitizing temperature is set

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from 850 °C to 1050 °C in the present study to investigate the resulting microstructure and mechanical properties. The investigation would deepen the understanding about ADI.

Based on the foregoing, the effects of austempering process parameters including austenitizing temperature as well as austempering temperature and time on the ADI microstructure and mechanical properties were evaluated systematically in this study, in order to provide a standard of reference for microstructure design and optimization of properties.

2. Experimental procedures

In this study, a hollow ductile iron pipe with a diameter of 100 mm and a thickness of 15 mm was fabricated via vertical continuous casting, and the detailed fabrication procedure has been reported previously [20]. The chemical composition of the ductile iron was: 3.4 wt% C, 2.4 wt% Si, 0.14 wt% Mn, 0.013 wt% S, 0.05 wt% P, 0.04 wt% Mg, and 0.03 wt% RE. The Fe level was balanced with a QSN-750 optical emission spectrometer. The microstructure of ductile iron is shown in Fig. S1 in our supplementary file.

The as-cast ductile iron was first heated to austenitizing temperatures (T_y) of 850 °C, 900 °C, 950 °C, 1000 °C, and 1050 °C for 90 min to obtain a full austenite matrix in a vacuum furnace. It was then rapidly quenched in a salt bath consisting of 50% NaNO₂ and 50% NaNO₃ at austempering temperatures (T_A) of 250 °C, 300 °C, and 350 °C and held for different austempering times (t_A) of 30 min, 60 min, and 90 min, respectively. Subsequently, the samples were air-cooled to ambient temperature. The samples prepared under different austempering process parameters were denoted as $T_y - T_A - t_A$. The single variable method was used to investigate the relationship among process parameters, microstructure, and mechanical properties of ADI, which was based on an initial austempering cycle of 900-300-60.

The samples intended for microstructure analysis were mechanically ground, polished, and etched with 4% Nital solution. The microstructure was analyzed using an OLYMPUS GX71 optical microscope (OM) and a JSM-6700F scanning electron microscope (SEM) with an accelerating voltage of 30 KV. The SEM images of ferrites were measured using Image J software. Further detailed microstructures were characterized via JEM-3010F transmission electron microscopy (TEM) with an accelerating voltage of 300 KV. The specimens used for TEM observation were first mechanically polished, followed by twin-jet electropolishing in a solution of 95% perchlorate and 5% alcohol, and then ion-milled to remove the oxide layer. In addition, X-ray diffraction (XRD) tests were used to estimate the volume fraction of austenite and its carbon content, using an XRD-7000 diffractometer with a scanning range of 30°–90° and a scan speed of 4°/min. A monochromatic Cu K α radiation with a wavelength (λ) of 1.5406 Å was used. The volume fraction of austenite in ADI was estimated via direct comparison by determining the peak intensities of (111) γ , (200) γ , (220) γ , (110) α , (200) α , and (211) α . The carbon content of austenite was calculated using the following equation [21]:

$$a_\gamma = 0.3548 + 0.0044 C_\gamma \quad (1)$$

where a_γ represents the lattice parameters of austenite (nm), which were obtained via Nelson–Riley extrapolation [22]; C_γ represents the carbon concentration of austenite (wt.%). In order to investigate the microstructure evolution, the MUCG83 thermodynamic model [23] based on the simulated elements equilibrium composition [2] was applied to calculate the nucleation driving forces of ferrite during austempering process.

The samples with a cross-section of 4 mm × 2 mm² and a gauge length of 20 mm were subjected to tensile tests using an HT-2402 universal testing machine at room temperature with a crosshead speed of 1 mm/min. Unnotched samples measuring 10 mm × 10 mm × 55 mm were used for Charpy impact tests using a JBN-300B impact testing

machine, according to the ASTM E–23 standard. Tensile and impact tests were conducted at least three times to ensure repeatability. The fracture morphology was analyzed via SEM. Hardness tests were carried out using an HVT-1000 microhardness tester with a load of 100 g and a holding time of 15 s. At least 10 points were obtained and the average value was determined.

3. Results

3.1. Microstructure

Fig. 1 shows the SEM images of ADI samples. A typical microstructure of ADI composed of spheroidal graphite, acicular ferrite, and high-carbon austenite could be seen. The SEM images of samples austenitized at 900 °C and 1050 °C are presented in Fig. 1. The SEM images of samples austenitized at 850 °C, 950 °C, and 1000 °C are shown in Fig. S2 of our supplementary materials. Significant differences in the length and thickness of acicular ferrite and the amounts of high-carbon austenite indicated that austempering process parameters had a significant effect on the ADI microstructure. The length of acicular ferrite was increased with increasing austenitizing and austempering temperatures, while the thickness showed the opposite trend. Under prolonged austempering, the dimensions of acicular ferrite changed little. Additionally, the volume fraction of austenite was increased with increasing austenitizing temperature (Fig. 1b) and austempering temperature (Fig. 1d) but reduced with increasing austempering time (Fig. 1f).

The variations in the dimensions of acicular ferrite are presented in Fig. 2. The results of the frequency distribution are presented in Fig. S3 of our supplementary file. It could be seen that the length of acicular ferrite was increased with increasing austenitizing temperature, while the thickness showed the opposite trend. The thickness of acicular ferrite decreased from 0.18 μ m for sample with 850-300-60 cycle to 0.04 μ m for sample with 1050-300-60 cycle, whereas the length increased from 4.23 μ m to 12.8 μ m. However, the length and thickness of acicular ferrite were increased with increasing austempering temperature. The ferrite thickness and length increased from 0.12 μ m to 1.16 μ m for sample with 900-250-60 cycle to about 0.24 μ m and 2.22 μ m for sample with 900-350-60 cycle. When the austempering time was extended, the acicular ferrite length decreased, while the thickness increased only slightly. Fig. 2 also shows the size variations of blocky austenite, which refers to sub-micron-sized austenite trapped between ferrites [24]. The size of blocky austenite increased with the increase in austenitizing and austempering temperatures, but decreased slightly with the increase in austempering time.

Fig. 3a–c shows the TEM images of ADI samples austempered at different temperatures. All samples exhibited a typical lamellar microstructure. The selected area of electron diffraction indicated that the lamellar microstructure consisted of acicular ferrite (α) and high-carbon austenite (γ) (Fig. 3b). Further, no carbides were detected because silicon atoms in ADI samples inhibited the precipitation of carbides [25]. Thus, the austempering temperature had a significant effect on the thickness of acicular ferrite, which increased from 0.06 μ m to 0.2 μ m when the austempering temperature increased from 250 °C to 350 °C. The variation was consistent with SEM results, but the values were significantly low mainly due to the differences in magnification. Fig. 3d shows a TEM image of ADI sample with 1000-300-60 cycle. The thickness of the ferrite decreased from about 0.08 μ m to 0.03 μ m (Fig. 3b) when the austenitizing temperature increased from 900 °C to 1000 °C. Additionally, the TEM images of ADI austempered at 300 °C and held for different austempering times are presented in Fig. S4. The results showed minor changes in the ferrite dimension with increasing austempering time.

Fig. 4 illustrates the XRD patterns of ADI samples. The peaks of α and γ phases were observed. However, the relative intensity and diffraction angles of γ phase peaks differed, indicating changes in the volume fraction of austenite and its carbon content under different

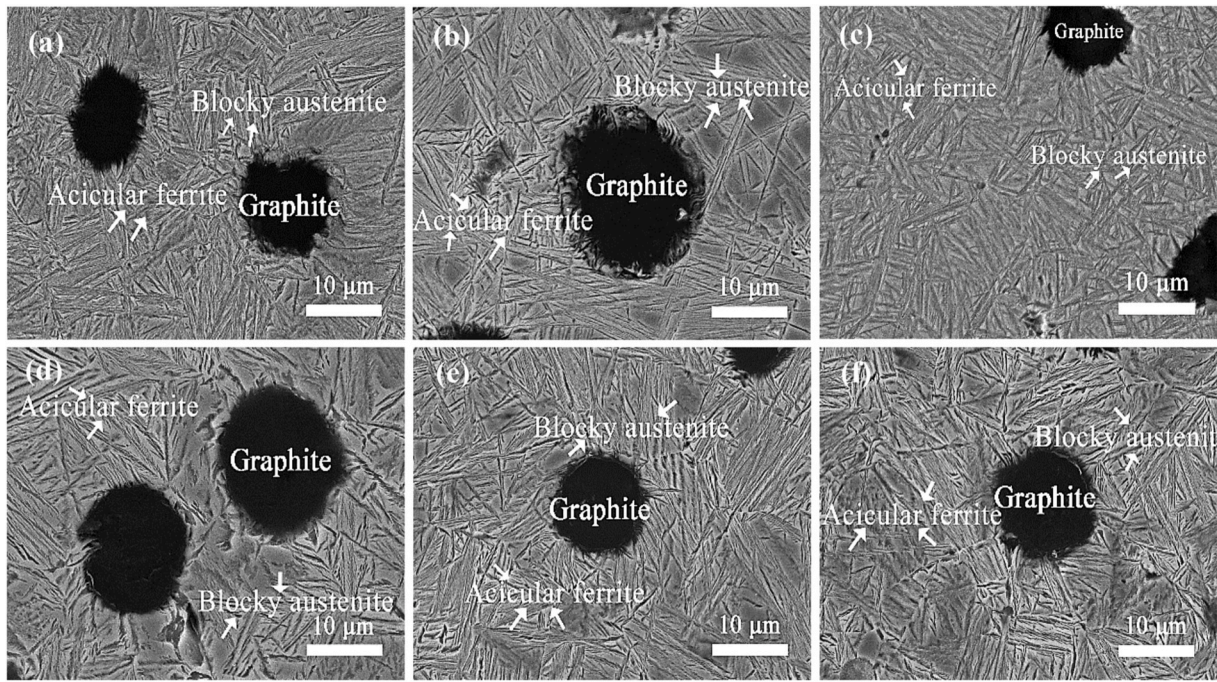


Fig. 1. SEM images of ADI samples under different austempering processes: (a) 900-300-60, (b) 1050-300-60, (c) 900-250-60, (d) 900-350-60, (e) 900-300-30, and (f) 900-300-90.

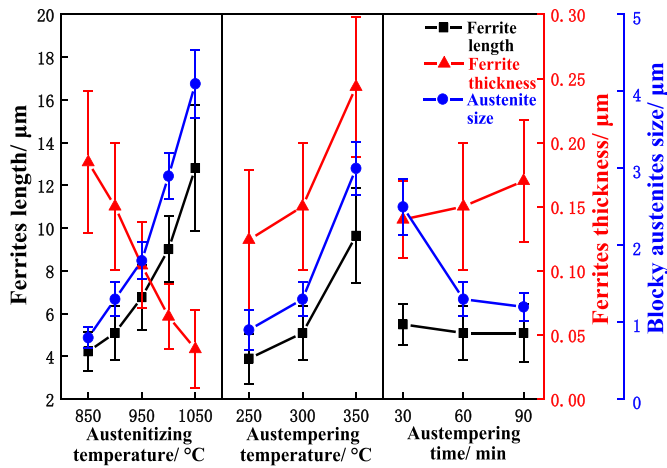


Fig. 2. Variations of ferrite dimension and blocky austenite size of ADI samples under different austempering processes.

austempering parameters.

The volume fraction of austenite and its carbon content was calculated. As shown in Fig. 5, the volume fraction of austenite and its carbon content increased with austenitizing and austempering temperatures. High austenitizing temperature promoted the diffusion of carbon atoms from graphite to prior austenite during the austenitizing stage, which facilitated the stability of austenite. Additionally, a high austempering temperature enhanced the diffusion of carbon atoms from α to γ phase, which further stabilized the austenite and resulted in a high-volume fraction of austenite. With prolonging austempering time, additional ferrites were transformed from austenite and thus reduced the austenite levels.

3.2. Mechanical properties

Fig. 6 shows the typical tensile stress-strain curves of ADI samples

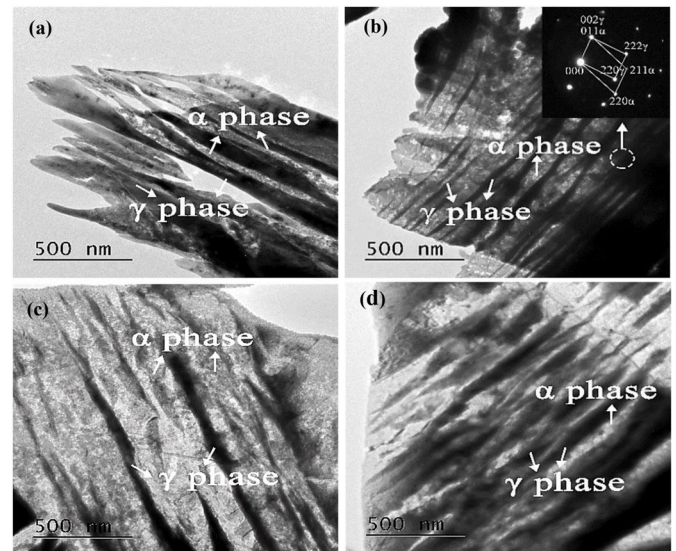


Fig. 3. TEM images of ADI samples under different austempering processes: (a) 900-250-60, (b) 900-300-60, (c) 900-350-60, and (d) 1000-300-60.

under different austempering parameters. The variations in yield stress (YS) and ultimate tensile strength (UTS) obtained from stress-strain curves are presented in Fig. 7. The variations in hardness are also presented in Fig. 7, which showed a similar tendency with strength. With increasing austenitizing temperature, the strength was first increased and then decreased. The UTS increased from 1254 MPa for sample with 850-300-60 cycle to 1348 MPa for sample with 900-300-60 cycle and then decreased to 1036 MPa for sample with 1050-300-60 cycle. In addition, the strength was decreased with the increase in austempering temperature. The UTS of the sample austempered at 250 °C decreased from 1417 MPa to 1058 MPa for the sample austempered at 350 °C. With the increase in the austempering time from 30 min to 60 min, the strength was increased. Further extending the holding time to 90 min led

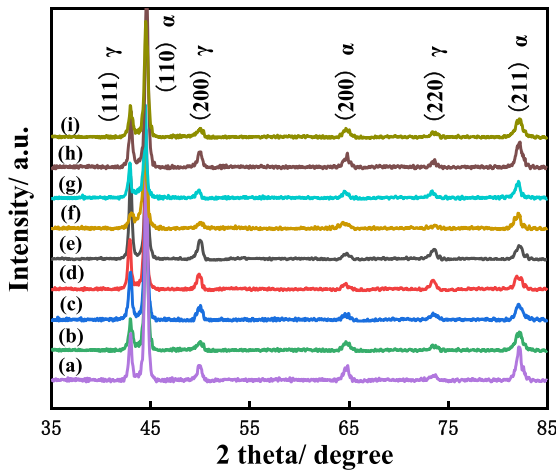


Fig. 4. XRD patterns of ADI samples under different austempering processes: (a) 850-300-60, (b) 900-300-60, (c) 950-300-60, (d) 1000-300-60, (e) 1050-300-60, (f) 900-250-60, (g) 900-350-60, (h) 900-300-30, and (i) 900-300-90.

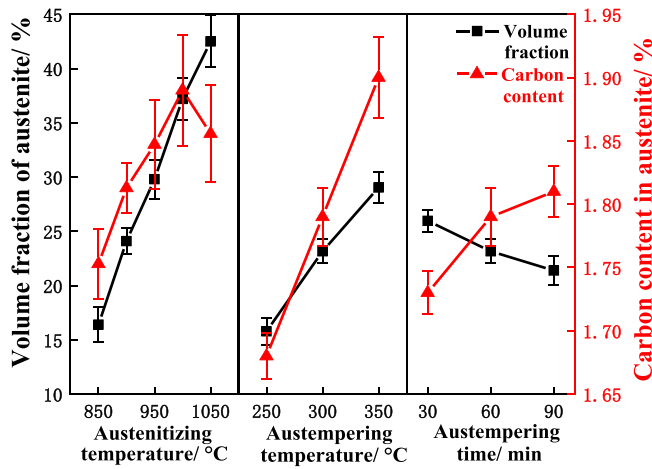


Fig. 5. Variations of the austenite volume fraction and its carbon content of ADI samples under different austempering processes.

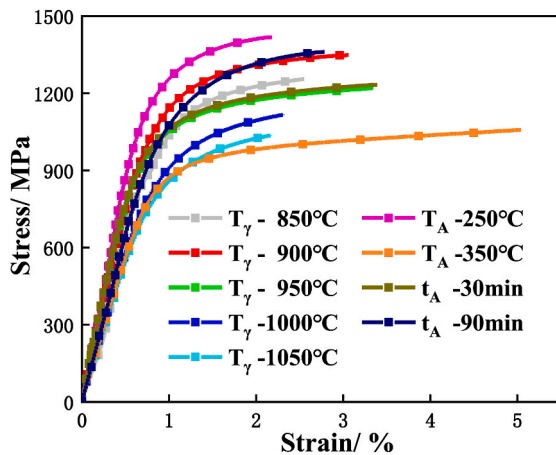


Fig. 6. Typical stress-strain curves of ADI samples under different austempering processes.

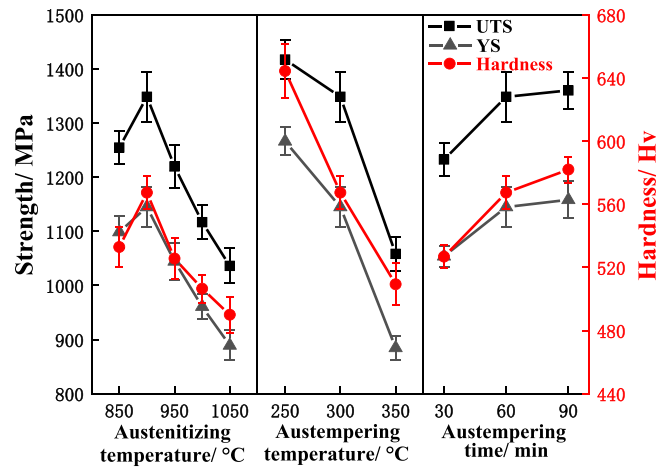


Fig. 7. Variations of the strength and hardness of ADI samples under different austempering processes.

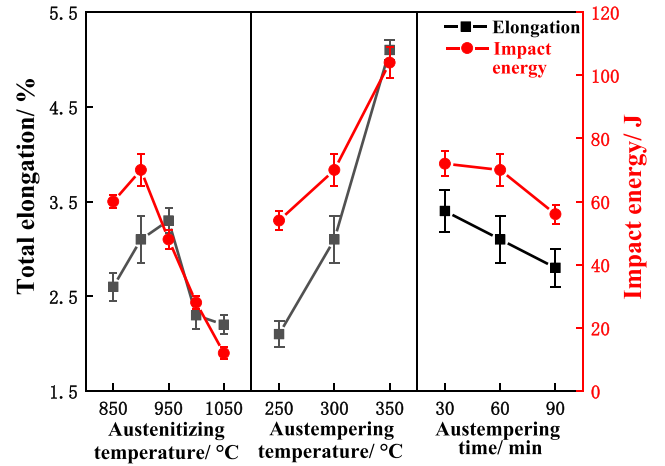


Fig. 8. Variations of the elongation and impact energy of ADI samples under different austempering processes.

to a minimal increase in strength, indicating cessation of α/γ transition.

Fig. 8 shows the variations in impact energy and total elongation (TL) of ADI samples under different austempering parameters. With increasing austenitizing temperature, the ductility of ADI first increased and then decreased, similar to strength variation. The TL of ADI sample with 850-300-60 cycle increased from 2.6% to 3.4% when it was austenitized at 950 °C and then decreased to 2.2% at 1050 °C. Additionally, with the increase in austempering temperature, the TL of ADI increased rapidly, contrary to strength variation. It increased from 2.1% for samples with 900-250-60 cycle to 5.1% for samples with 900-350-60 cycle. With the extension of austempering time, the γ phase transformed into the α phase, thus decreasing its elongation accordingly.

Impact energy variations exhibited the same trend as the elongation, but with minor differences. With the increase in austenitizing temperature, the mechanical properties of ADI deteriorated, which led to a deterioration of toughness. When the austenitizing temperature exceeded 950 °C, the impact toughness worsened seriously. The impact energy decreased from 70 J for sample with 900-300-60 cycle to 12 J with 1050-300-60 cycle. Additionally, when the austenitizing temperature was increased from 900 °C to 950 °C, although elongation was improved, the impact energy decreased because of the strength deterioration. Impact toughness was the comprehensive response to strength and ductility [26]. The impact energy increased with the increase in

austempering temperature. ADI sample with 1050-300-60 cycle showed maximum impact energy of 104 J, which was attributed to its strong ductility. Additionally, when the austempering time was increased from 30 min to 60 min, the increased strength offset the reduction of ductility, resulting in little change in impact energy. However, further prolongation of austempering led to the loss of ductility and reduced the impact toughness.

4. Discussion

The effects of austempering parameters on the microstructure and mechanical properties of ADI were investigated in this study. The fraction of α/γ phases and the size of acicular ferrite are closely related to heat treatment parameters. Consequently, the samples fabricated under different heat treatment parameters exhibited different mechanical responses. The microstructure evolution and strengthening mechanisms are discussed below.

4.1. Microstructure evolution of ADI

The matrix of ADI sample was composed of acicular ferrite and high-carbon austenite. Acicular ferrites were nucleated first at the interface of austenite/graphite and prior austenite grain boundaries, followed by nucleation at these newly created nucleation sites [27]. The nucleation driving force of the ferrite nucleus reduced free energy [28], which was calculated using the MUCG83 thermodynamic model, as shown in Fig. 9. Increasing austenitizing and austempering temperatures reduced the nucleation driving force and resulted in a low nucleation rate. The length of ferrite was affected by the nucleation driving force. Therefore, at low austenitizing and austempering temperatures, the high nucleation driving force facilitated the formation of ferrite nucleus, which interfered with each other during the subsequent growth and resulted in the formation of small acicular ferrites (Fig. 1a and c). Conversely, fewer ferrite nuclei were formed at high austenitizing and austempering temperatures. Consequently, the unhindered acicular ferrite traversed the austenite grain and exhibited a long length (Fig. 1b and d) [29]. A similar microstructure evolution has been reported in bainitic steels [28]. Additionally, the prior austenite grain was larger at high austenitizing temperatures, resulting in longer acicular ferrites.

The thickness of acicular ferrite depended primarily on austenite strength and nucleation rate [30]. With increasing austenitizing temperature, the carbon content in prior austenite increased [19]. As a result, high carbon content improved the strength of prior austenite via

solution strengthening and limited the ferrite thickness. For example, the samples with 900-300-60 and 1050-300-60 cycles showed similar nucleation driving force, but the sample austenitized at 1050 °C exhibited a thinner ferrite (Fig. 1b) because the prior austenite containing additional carbon atoms limited the coarsening of ferrite. In the case of ADI samples under different austempering temperatures, the prior austenite strengths were similar and the ferrite thickness mainly depended on the nucleation rate. Therefore, both the ferrite length and thickness were increased with increasing austempering temperature (Fig. 1d).

The nucleation and growth of ferrite were an ongoing process. Prolonged austempering increased the formation of secondary ferrites, which was also reflected in the decrease of the austenite volume fraction (Fig. 5). The growth of secondary ferrites (nucleated via autocatalytic nucleation) was blocked by the preformed ferrites (nucleated via grain-boundary nucleation), resulting in reduced length [31]. For ADI austenitized at 900 °C and austempered at 300 °C, when austempering time exceeded 60 min, the untransformed austenite was stabilized due to the increased carbon concentration [32]. Although the total amount of acicular ferrite increased, the transformation rate was gradually reduced, resulting in minimal changes in the average thickness of ferrite (Fig. 1f).

Carbon atoms diffused from acicular ferrite to surrounding austenite and increased the stability of austenite during the ferrite growth [33]. The ferrites showed consistent orientation and were separated by thin filmy austenite, which formed ausferrites. The regions trapped with ausferrites formed blocky austenite. At low austempering and austenitizing temperatures, the dense ferrites resulted in smaller blocky austenites measuring less than 0.1 μm (Fig. 2). Low nucleation rates at high austenitizing and austempering temperatures enabled ausferrites to traverse austenites without interference and resulted in large-scale blocky austenites (Fig. 2), such as the sample austempered at 350 °C ($\sim 0.2 \mu\text{m}$). Similar results reported by others [17,31]. Notably, with increasing austenitizing temperature, the prior austenite carried increased carbon content and showed limited carbon diffusion from ferrite. Thus, the samples austenitized at 1050 °C showed a microstructure characterized by large-scale high-carbon austenite with slender traversing ferrites.

4.2. Mechanical properties analysis of ADI

The samples under different austempering parameters exhibited different mechanical properties. Strength and ductility are two conflicting mechanical properties. However, the strength and ductility of ADI decreased simultaneously when the austenitizing temperature exceeded 950 °C in the present study, because both the austenite and acicular ferrite were coarse, which greatly deteriorated the mechanical properties [34]. With increasing austempering temperature, the increased austenite fraction showed increased ductility but deteriorated strength. The ductility of ADI, which contained the same matrix as bainitic steel, was mainly controlled by the austenite fraction [35]. Further, the presence of fine ferrites improved strength significantly at low austempering temperatures, according to the Hall-Petch relationship [36]. When the austempering time was less than 60 min, the γ/α transition was incomplete and resulted in low strength. However, when the austempering was prolonged to 90 min, the γ/α transition was hindered and therefore resulted in similar mechanical properties.

Austenite is an important phase in improving ductility via transformation-induced plasticity (TRIP) [37]. In order to investigate the role of austenite in mechanical properties, the regions near the fracture surface of the tensile sample were analyzed by XRD. The selected areas in the tensile samples were prepared according to the scheme shown in Fig. 10. The variations in austenite volume fraction before and after tensile tests are also presented in Fig. 10, which revealed a slight decrease in the austenite volume fraction. The variation in the volume fraction of austenite in sample with 900-300-60 cycle

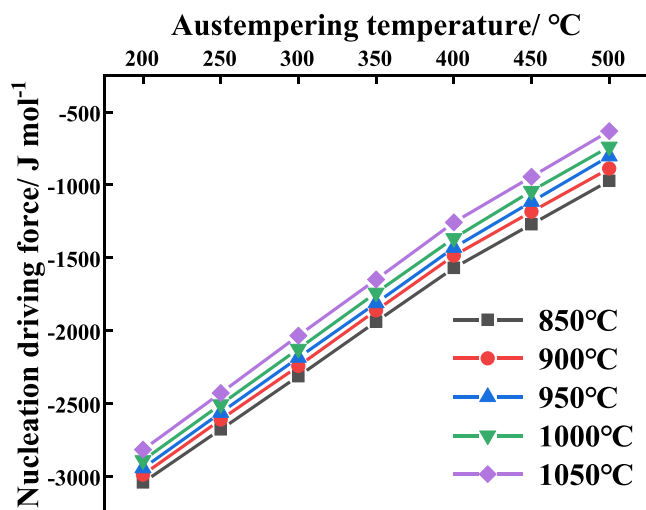


Fig. 9. Calculated nucleation driving force of ferrite in ADI under different austempering processes using the MUCG 83 code.

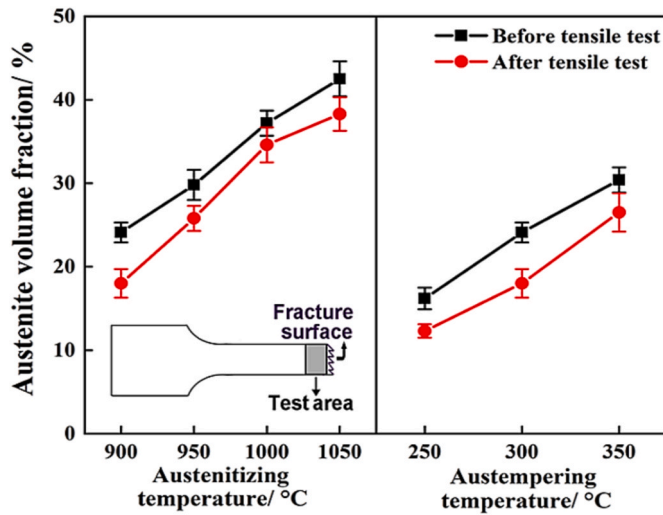


Fig. 10. Variations of the austenite volume fraction before and after tensile tests.

showed the largest reduction, but the value was only about 6%. It indicated that only a small amount of austenite was transformed during deformation because the high carbon concentration (>1.6 wt%) in the high-carbon austenite increased the mechanical stability and interfered with transformation [38]. Therefore, the phase transformation in ADI had little effect on the mechanical response.

Fig. 11 shows the fracture surfaces of ADI samples under different austempering parameters. Graphite and graphite cavities were observed on fracture surfaces. Moreover, large amounts of quasi-cleavage facets with few dimples were also found in ADI samples, indicating the brittleness of the samples. However, the proportions of dimples differed. The dimples decreased with increasing austenitizing temperature but increased with increasing austempering temperature. For example, the number of dimples in sample with 900-350-60 cycle (Fig. 11c) was more than that with 900-300-60 cycle (Fig. 11a). Additionally, the dimples decreased slightly with the increase in austempering temperature, as shown in Fig. S5. The fracture surfaces of ADI samples were consistent with the results of ductility. In addition, obvious cracks were detected in sample with 1050-300-60 cycle (Fig. 11b), because the oversized prior austenite grain showed deterioration in ductility, resulting in transgranular fracture.

Impact toughness is an important index for ADI. In the present study, impact toughness changed similarly to elongation (Fig. 8). However, the ductility was not the only factor determining impact toughness, which was affected by the overall mechanical properties. For example, the samples austenitized at 950 °C showed improved elongation, while their impact toughness was decreased due to the reduction in strength. In addition, a previous study showed that the impact toughness was

improved with the increase in austenite fraction [39]. However, ADI with 1050-300-60 cycle contained a large amount of austenite (>30%) but it showed a very low impact energy of 12 J because the previous study was based on the improvement of ductility by high volume fraction of austenite. High austenitizing temperature increased the volume fraction of austenite, but it also coarsened the austenite size and deteriorated the mechanical properties, which reduced the impact toughness significantly. The strength-ductility balance reflected the overall performance of the materials. The variations in impact energy with strength-ductility balance are presented in Fig. 12, suggesting that the impact energy was approximately proportional to the strength-ductility balance. The relationship between strength-ductility balance and impact energy was approximately follow the equation:

$$A_k = -36 + 25U_T \quad (2)$$

Where A_k is impact energy (J) and U_T is strength ductility balance (GPa %). Previous results presented a proportional relationship with different slopes [18,40], which was because the strength-ductility balance of ADI depended on chemical composition, casting, and experimental environment [41].

In the present study, even under different austempering parameters, the maximum strength difference of ADI was less than 400 MPa, whereas the maximum elongation was 2.3-fold higher than the minimum elongation. Therefore, the impact toughness of ADI was more sensitive to ductility. For example, the sample with 900-250-60 cycle having high strength (1417 MPa) and low elongation (2.1%) showed a medium impact energy (54 J). However, the sample with 900-350-60 cycle having low strength (1058 MPa) and high elongation (5.1%) exhibited high impact energy (104 J), which was 2-fold higher than that of the samples austempered at 250 °C.

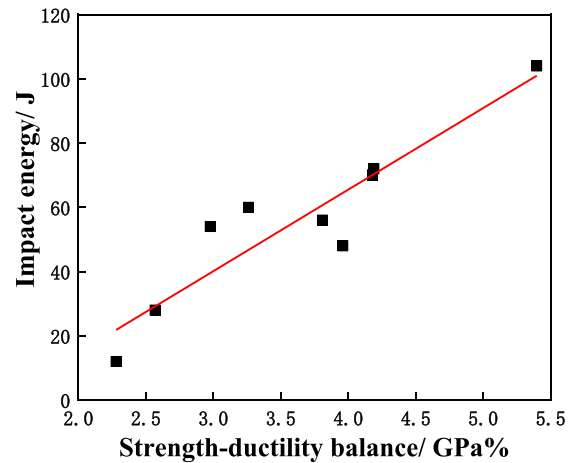


Fig. 12. Variations of impact energy with strength-ductility balance.

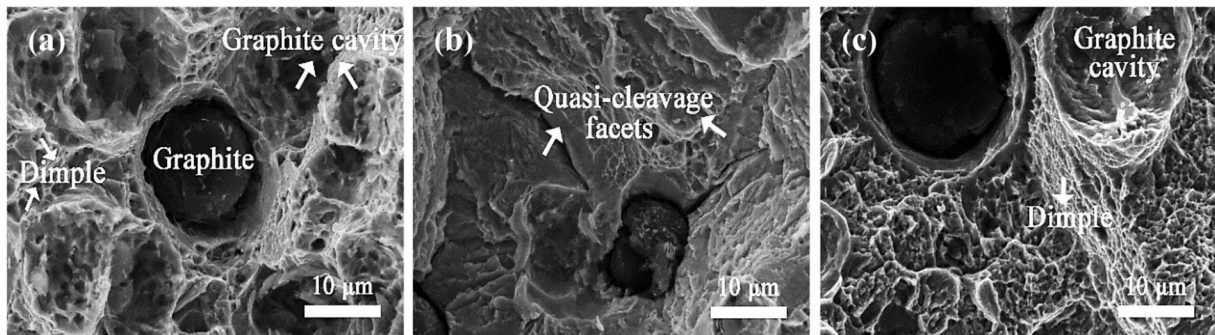


Fig. 11. Fracture surfaces of ADI samples under different austempering processes: (a) 900-300-60, (b) 1050-300-60, and (c) 900-350-60.

In the current study, the microstructure and mechanical properties of ADI generated under different austempering cycles were investigated. The results showed that the austempering process parameters had a vital effect on the microstructure and final mechanical properties of ADI. Austenitizing temperature determined the matrix carbon content, which directly affected the dimensions and fractions of acicular ferrite. However, when the austenitizing temperature exceeded 950 °C, the coarsening of the prior austenite deteriorated the mechanical properties significantly. Additionally, the microstructure was coarsened with the increase in austempering temperature, which improved ductility but sacrificed strength. When the austempering time was less than 60 min, the α/γ transition was incomplete and interfered with the mechanical properties. Adjusting austempering process parameters to obtain a uniform and fine microstructure was the key to obtaining high-performance ADI.

5. Conclusions

The relationship among austempering process parameters, microstructure, and mechanical properties of ADI was investigated in the present study. The microstructure of ADI was sensitive to process parameters, which resulted in different mechanical responses. The main conclusions are summarized below.

- (1) Austenitizing temperature affects the microstructure and mechanical properties of ADI. Increased austenitizing temperature results in an increase in the length of acicular ferrite, while the thickness is reduced and the volume fraction of high-carbon austenite is increased. The strength of ADI is first increased and then decreased with the increase in austenitizing temperature, which is related to the size of acicular ferrite and relative amounts of austenite. When the austenitizing temperature is 900 °C, ADI exhibit a good combination of mechanical properties with a UTS of 1348 MPa, TL of 3.1%, and impact energy of 70 J.
- (2) The size of acicular ferrite and the fraction of high-carbon austenite are increased with the increase in austempering temperature. Consequently, the strength of ADI is decreased, while the ductility is increased under increased austempering temperature. The UTS of the sample austempered at 250 °C decreases from 1417 MPa to 1058 MPa for the sample austempered at 350 °C, while the ductility increases from 2.1% to 5.1%, nearly 2.5 times.
- (3) The degree of transformation from prior austenite to acicular ferrite is increased with the increase in austempering time. However, further extending the austempering time to 90 min, the microstructure and mechanical properties of ADI change little.
- (4) Impact toughness of ADI is determined by strength and ductility, which is approximately proportional to strength-ductility balance.

CRedit authorship contribution statement

Xin Wang: Investigation, Data curation, Writing – original draft. **Yuzhou Du:** Writing – review & editing, Conceptualization, Funding acquisition, Formal analysis, Supervision. **Chen Liu:** Investigation, Writing – review & editing, Supervision. **Zhitao Hu:** Methodology, Resources. **Zhijie Gao:** Investigation, Formal analysis, Validation. **Hui Guo:** Methodology, Resources, Formal analysis. **Bailing Jiang:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.msea.2022.144063>.

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